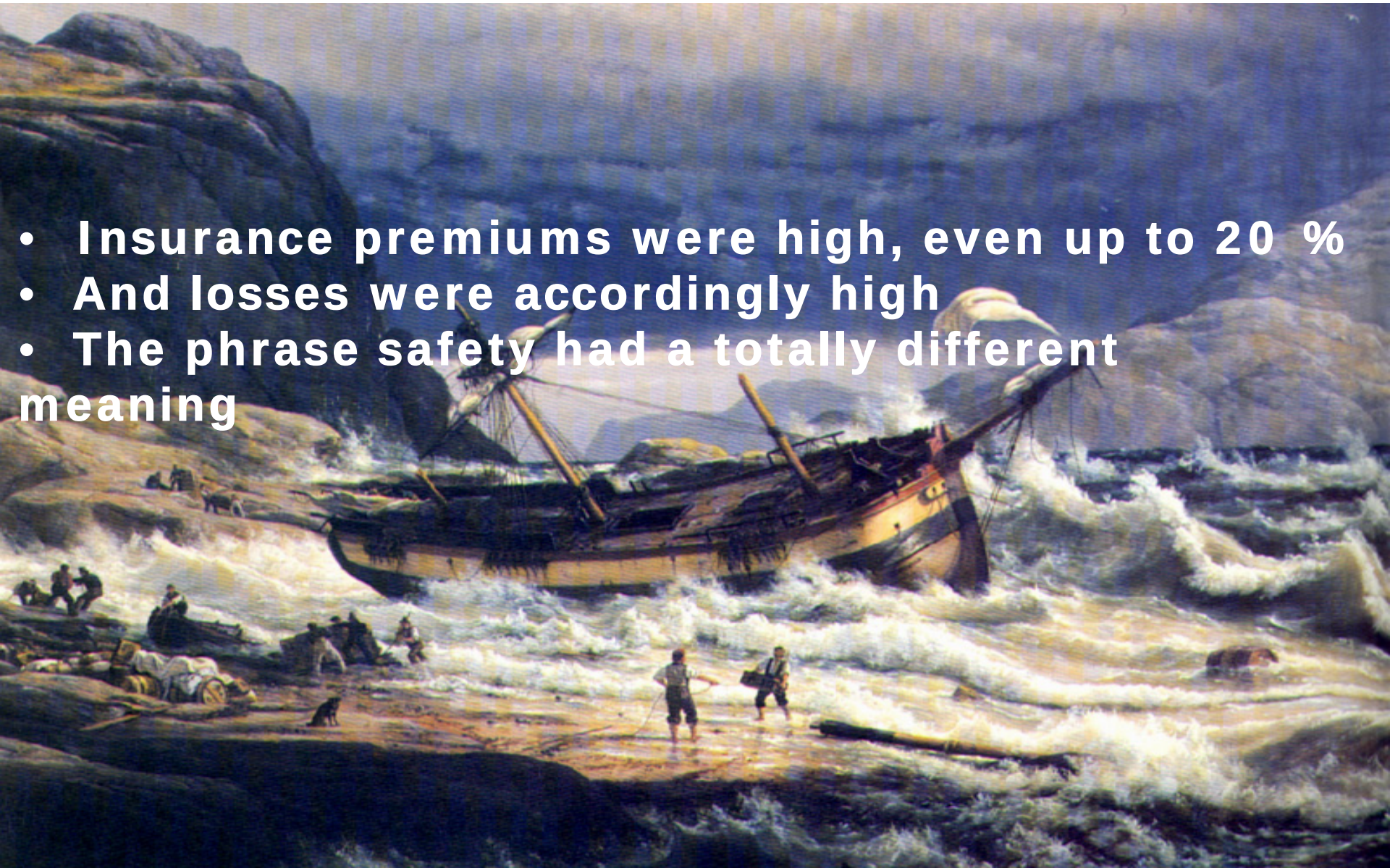
THE CLASSIFICATION CONCEPT

Jonas Mattos

02 September 2016

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Sea transportation - mid 19th century.



- Insurance premiums were high, even up to 20 %
- And losses were accordingly high
- The phrase safety had a totally different meaning

Classification Societies – Background

- A short history of Classification & statutory roles in the context of the times
- Increase in international **trade** in the middle of the 18th Century
- **Ship owners** needed insurance to reduce their economical risk
- **Insurers** needed someone to establish safety standards for ships and to verify that the standards were complied with
- Initially it was all about property



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Origin of Classification

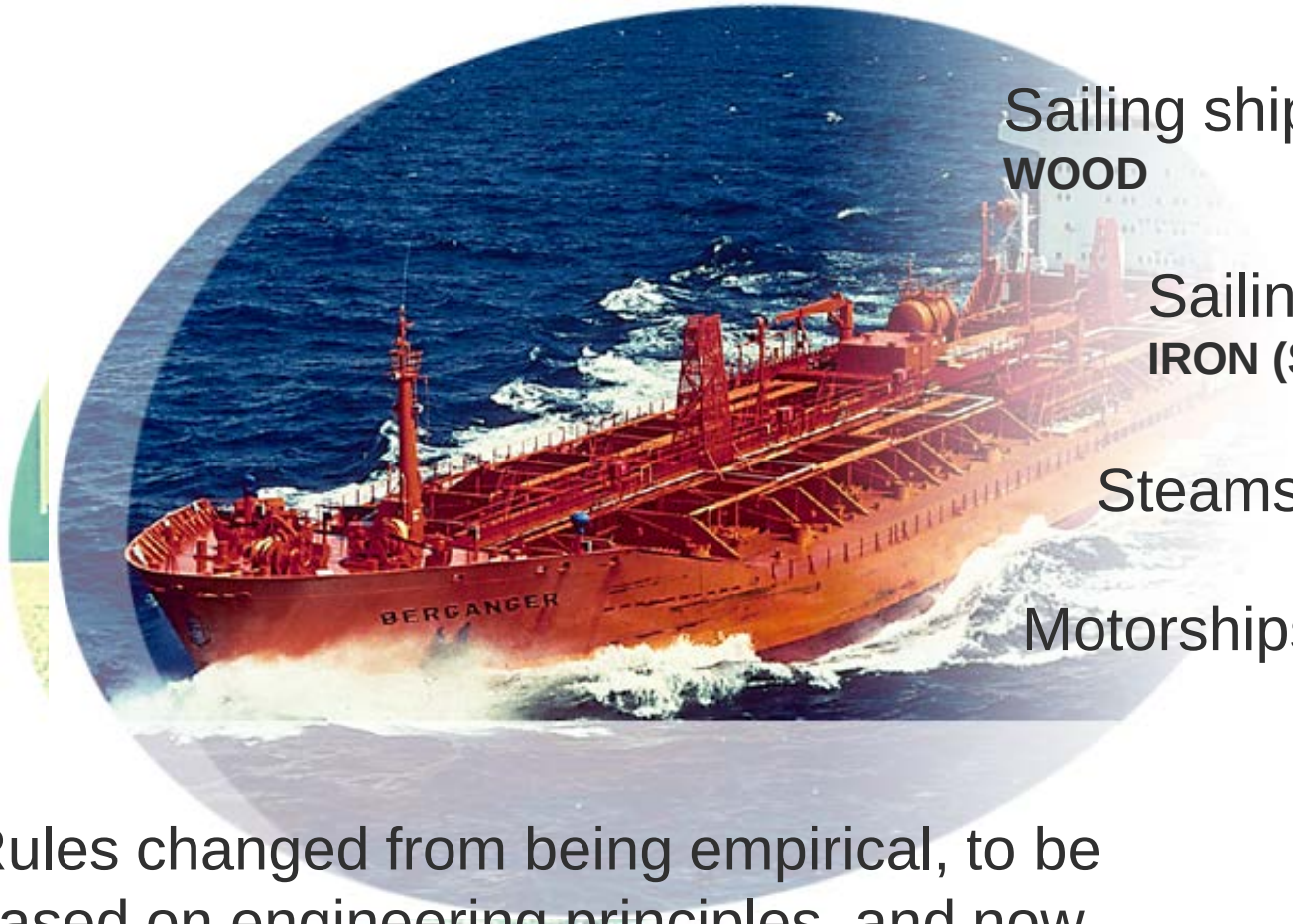


OWNER
2nd party

CLASSIFICATION
SOCIETY
3rd party

INSURER/
UNDERWRITER
1st party

Rules development



Sailing ships
WOOD

Sailing ships
IRON (STEEL)

Steamships

Motorships

Rules changed from being empirical, to be based on engineering principles, and now also including risk assessment options.

Classification Societies

- The Rules:
- The Rules are technical standards based upon research, risk analysis and practical experience.
- DNV verify that these Rules are satisfied during design , construction and installation
- DNV verify that the standard of the ship and its equipment are acceptable and according to the Rules throughout its operational life through periodical surveys.



Who are Class' "main relations"?



IMO

IACS

INTERNATIONAL ASSOCIATION OF CLASSIFICATION SOCIETIES

DNV·GL

Re. Statutory regulations:
The Flag states

Re. Class rules
DNVGL

DNV·GL

Customers:

In construction phase: The Yard

In the operational phase. The Owner

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Background Statutory Requirements

"...in all my experience, I have never been in an accident of any sort worth speaking about.
I never saw a wreck and have never been wrecked, nor was I ever in any predicament that threatened to end in disaster of any sort."

interview in the New York press, 1907.

Edward J. Smith 1850 – 1912
Captain in command of the RMS *Titanic*



Accidents have always influenced rules and regulations

Examples:
Titanic (1912)
SOLAS (first edition)





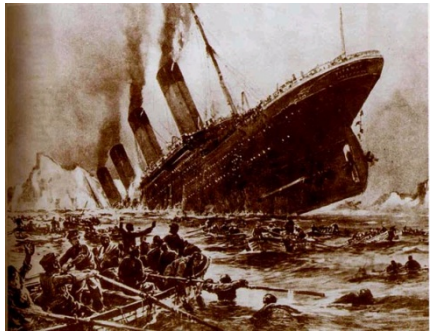
Torrey Canyon (1967)
MARPOL 1973

Herald of Free Enterprise (1987)
SOLAS ‘90’, ISM



Regulations in shipping tend to be ‘disaster driven’







Titanic (1912)	SOLAS (1929)
Torrey Canyon (1967)	MARPOL (1973) / STCW (1978)
Amoco Cadiz (1978)	SOLAS / MARPOL 1978 Protocols
Herald of Free Enterprise (1987)	ISM / SOLAS Ch. II-1 / FSA
Exxon Valdez (1989)	OPA 90 / MARPOL
Scandinavian Star (1990)	SOLAS Ch. II-2
Carriers lost early 1990	SOLAS Ch. XII (1997)
Estonia (1994)	SOLAS Ch. II-1 (1995)
Erika (1999)	Erika Pack I/II -> PSC/Class/EMSA
Prestige (2002)	Erika Package III>Flag State/Pollution Liability
09. 11(2002)	ISPS Code
Star Princess (2006)	SOLAS Ch. II-2
General latest years	Emission Controlled Areas, Discharge to Sea, Green Passport







Class Rules vs. International Legislation

International Legislation covers aspects related to :
Safety, Health and Environment

Accommodation
Life saving
Navigation
Fire
Load Line, Stability
Radio Communication
Dangerous goods
Security
Pollution Prevention
Technical safety (Class certificate)
Manning ...



Traditional Classification coverage

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International Maritime Organisation (IMO)

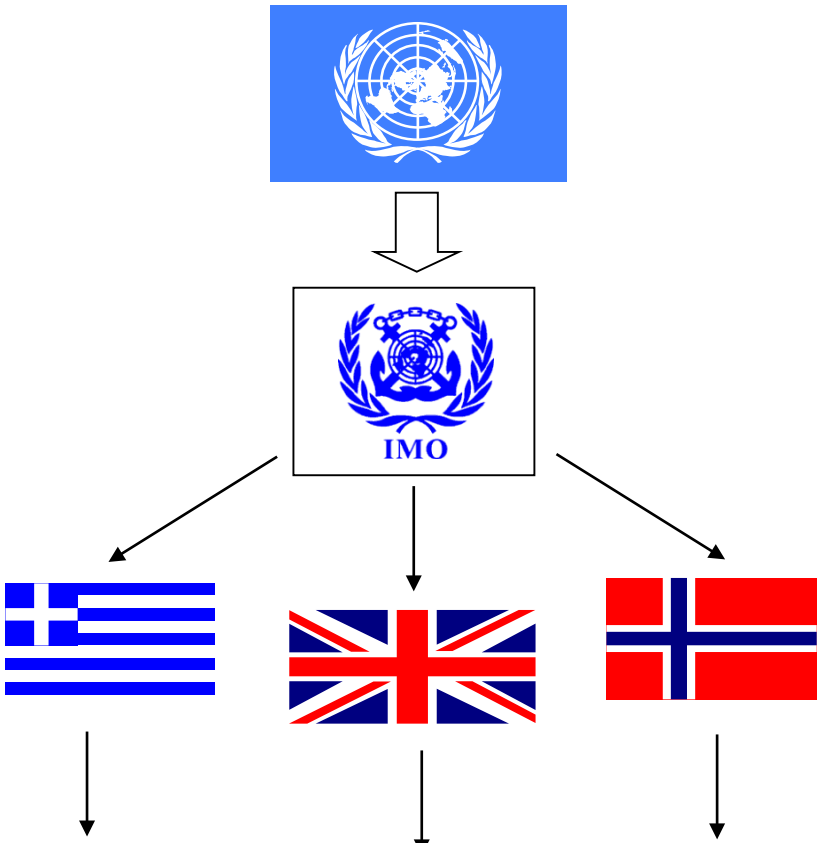


- United Nation's body for maritime affairs
- Develop and maintain conventions giving safety regulations for ships trading internationally
- Abt. 156 nations are members of IMO
- Secretariat is and meetings take place in London

- IMO has no operational role!

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The IMO - Flag state hierarchy



IMO
Conventions

Ratified by countries administrations to become National laws and regulations







Flag States

- A ship has to comply with the safety regulations issued by the Maritime Authority of the country whose flag the ship is flying (the Flag State)

- Countries are adopting and implementing the safety regulations given in the conventions, issued by IMO.

- The Maritime Safety Regime is a Flag State Regime as the Flag State is enforcing internationally agreed safety regulations (the conventions) through national maritime law.

Port States

- Verify compliance with international regulations for vessels entering their jurisdiction

- US Coast Guard
- Paris MoU
- Tokyo MoU



Main Class and additional class notations

Example: “HÖEGH TROOPER”:

Main Class: ✠1A1

Additional Class: ICE-C RO/RO MCDK EO NAUT-OC SC

A typical AHTS will be entitled the following notations:

✠1A1 Offshore Service Vessel AHTS

or; ✠1A1 Offshore Service Vessel+ AHTS

A typical PSV will be entitled the following notations:

✠1A1 Offshore Service Vessel Supply

DNV currently has more than 300 different additional class notations. Additional class notations are used to describe special features related to design, equipment and operation of the vessel

Plan Approval



"Plans" means all documentation needed to construct a ship. Specified parts of the design documentation is submitted to DNV for "Plan approval", and are approved when it is verified that Rule requirements are met.

Important dates:

Date of contract between yard and owner:

Determines which edition of class rules to be used.

Date of keel laying:

Determines which statutory regulations which will apply

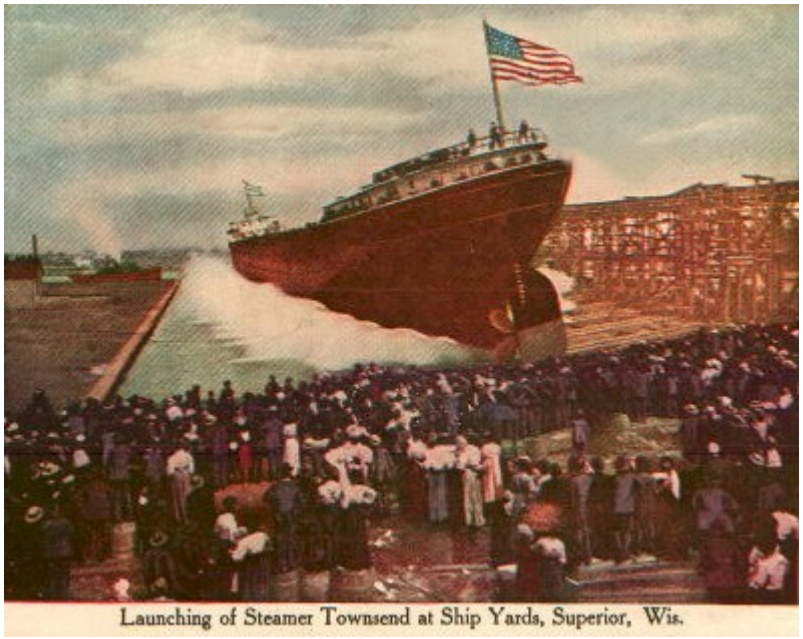
Survey during construction



The shipyard will construct the ship according to the approved design documentation. The local DNV office at the yard will follow up the construction.

Certification of materials and components will be verified by DNV office at the yard, or at other DNV offices where components are manufactured.

Delivery


Launching of Steamer Townsend at Ship Yards, Superior, Wis.

Upon delivery DNV changes contractual partner from yard to owner.

Interim Class and Short Term Statutory certificates are issued.
All documentation from the construction phase is submitted to
the Class main office for evaluation and filing, for re-use in the operational phase.

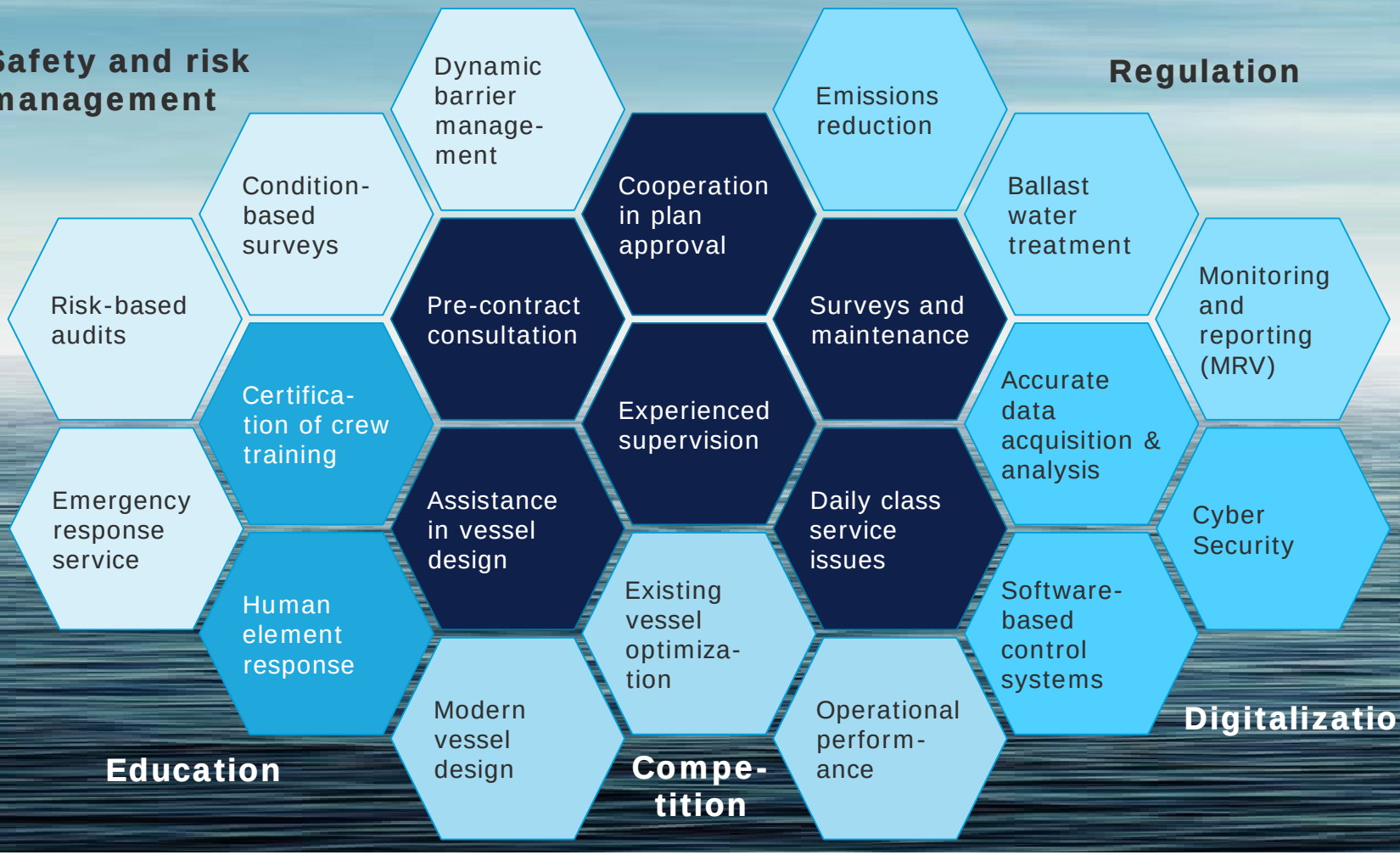
When documentation is found satisfactory, full term
Certificates are issued, valid for
5 years.

In a challenging shipping market, our core class services are more relevant than ever for maritime customers…

Newbuilding and Fleet in Service



… but the role of class has expanded significantly over the years as the needs of maritime customers have grown rapidly as well



The Human Factor



The Problem is not a lack of Rules and Regulation - but Compliance !

Thank you!

www.dnvgl.com

SAFER, SMARTER, GREENER